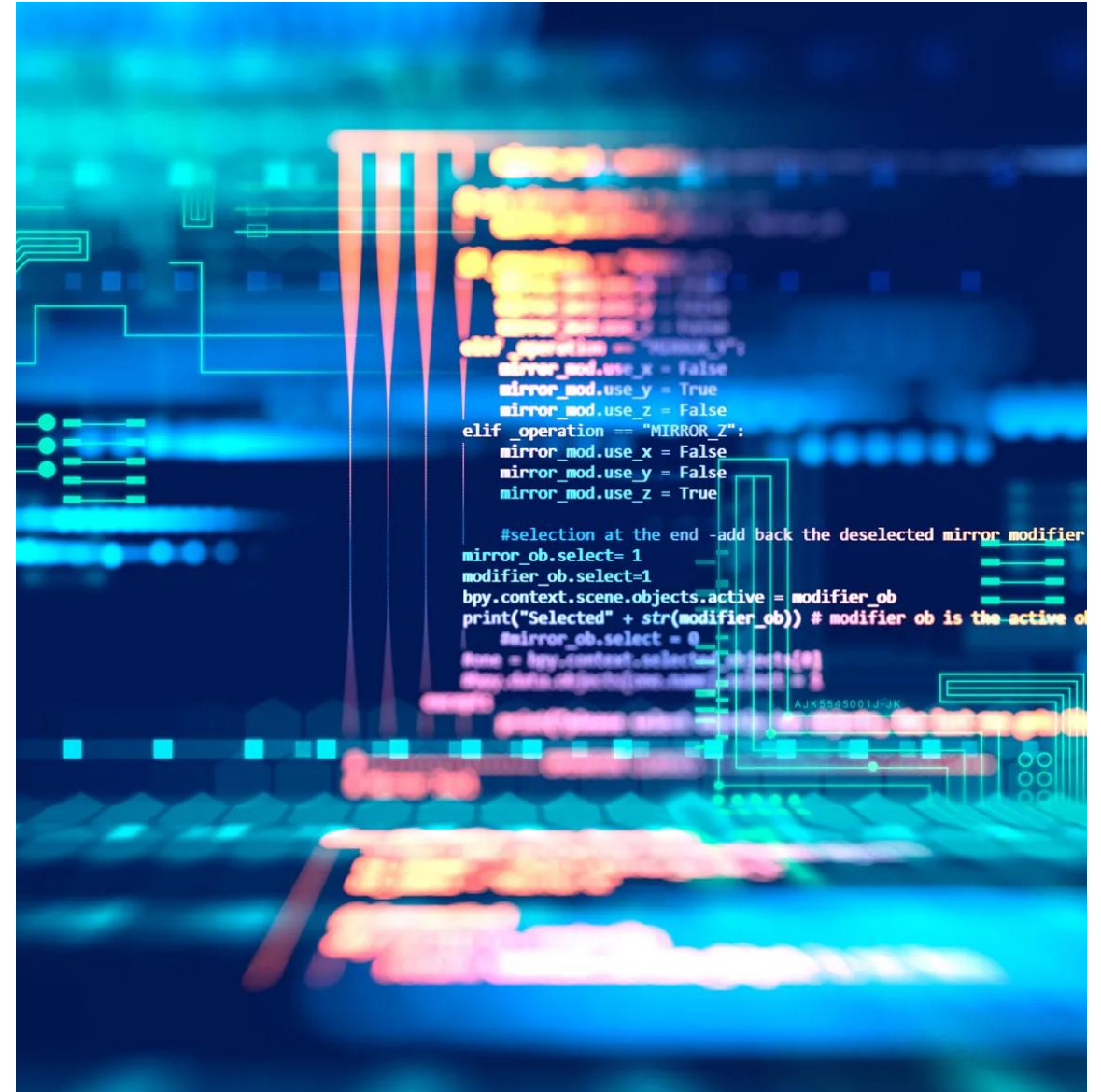


CS 410 CONCEPTS OF PROGRAMMING LANGUAGES

Presented By:

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LECTURE OBJECTIVES

- Introduction To Concepts of Programming Languages
- Course Objectives and outcomes
- Prerequisite courses
- Textbooks and Related references
- Communication Platform
- Grading Policies
- Summary

INTRODUCTION TO CONCEPTS OF PROGRAMMING LANGUAGES

- Concepts of Programming Languages is a course that helps to understand how different programming languages are built and how they work.
- Instead of focusing on just one language, it explains the common ideas that many languages share, such as how variables are stored, how programs are structured, and how instructions are executed.
- By learning these concepts, students can compare different languages, choose the right one for a problem, and become better programmers overall.

COURSE OBJECTIVES AND OUTCOMES

- The course aims to: .
 1. Introduce the main principles behind programming language design and use.
 2. Explain the similarities and differences among various programming paradigms (imperative, functional, object-oriented, logic).
 3. Help to understand how programming languages handle syntax, semantics, variables, and memory.
 4. Develop the ability to analyze and compare programming languages.
 5. Provide a foundation for advanced topics like compilers, software engineering, and systems programming.

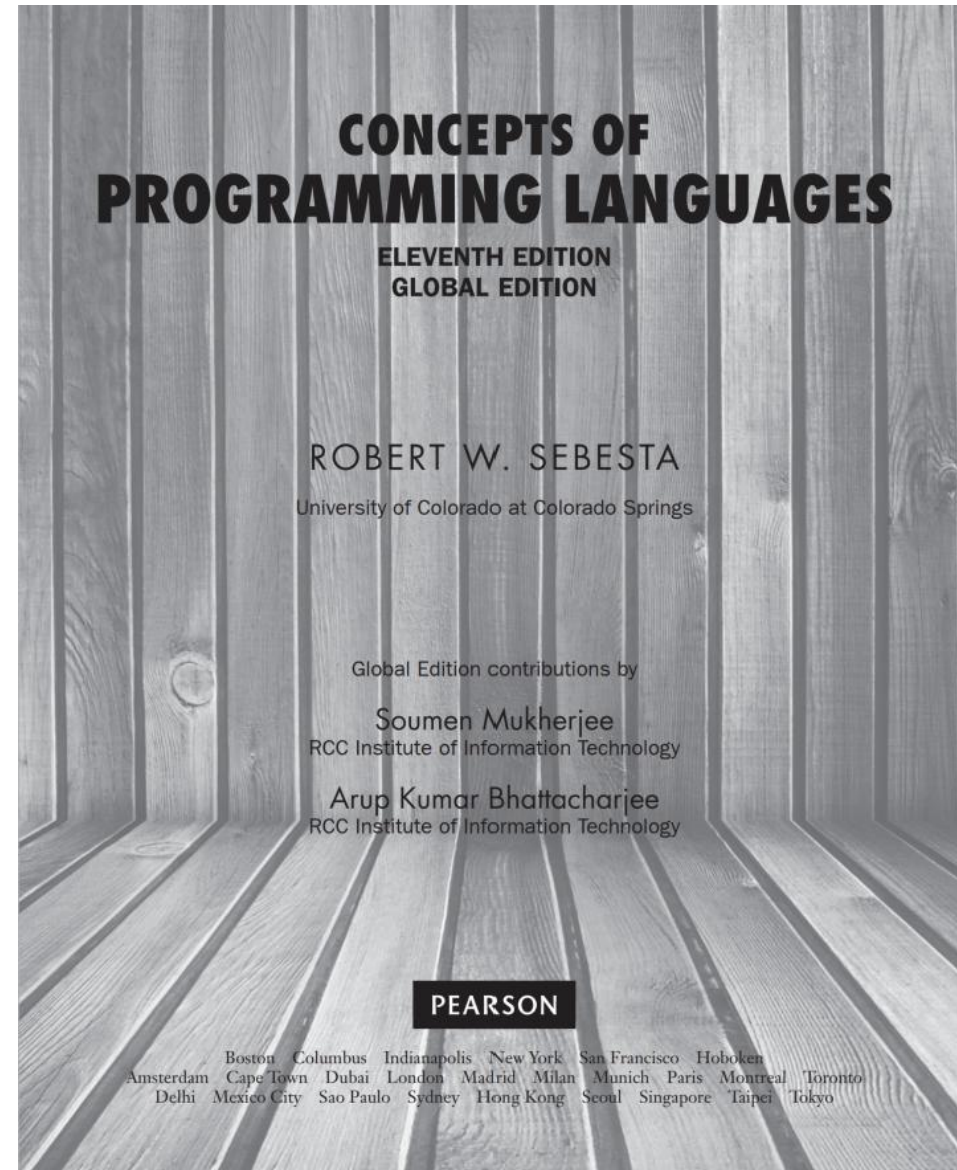
COURSE OBJECTIVES AND OUTCOMES

- The outcome for that course will be:
 1. Describe the basic concepts and features common to most programming languages.
 2. Explain different programming paradigms and when to use them.
 3. Understand how languages handle variables, data types, control flow, and memory.
 4. Compare and evaluate the strengths and weaknesses of different programming languages.
 5. Apply language concepts to solve computational problems more effectively.

PREREQUISITE COURSES

- Prerequisite course for Course will be mainly Programming Language 1
 1. Prior experience with at least one high-level programming language (e.g., Java, Python, C, or C++).
 2. Ability to write, debug, and understand simple programs.
 3. Understanding of fundamental programming constructs: variables, loops, functions, recursion, and data structures.
 4. Familiarity with problem-solving and algorithmic thinking.
 5. Understanding of logic, sets, and discrete mathematics concepts.

TEXTBOOKS AND RELATED REFERENCES



COMMUNICATION PLATFORM

- Official Communication Platform will be MS Teams
 1. **Channel Name:** CS 410 Concepts Of Programming Languages_Autumn_ 2025
 2. **Code:** ia173hu



GRADING POLICIES

- Grading Policies will be according to the following:
 1. 50% For Final Exam
 2. 30% For Midterm
 3. 20% Presentations and Tasks.

SUMMARY

Introduction: Understanding how different programming languages are built and work.

Course Objectives:

- Principles of programming language design and use.
- Similarities and differences among programming paradigms.
- Syntax, semantics, variables, and memory handling.
- Analyzing and comparing programming languages.
- Foundation for advanced topics like compilers and software engineering.

Outcomes:

- Describe basic concepts and features of programming languages.
- Explain different programming paradigms.
- Understand variables, data types, control flow, and memory.
- Compare and evaluate programming languages.
- Apply language concepts to solve problems.

Prerequisites: Prior experience with a high-level programming language (e.g., Java, Python, C, or C++).

Textbooks and References: Provided in the course.

Communication Platform: MS Teams.

Grading Policies: 50% Final Exam, 30% Midterm, 20% Presentations and Tasks.